

SE Lab-03

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Section:-K

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Topic:- Local Sports Tournament & Bracket Manager.

Architecture Selection

We chose Layered Architecture for the Local Sports Tournament & Bracket Manager.

Architectural Choice

Layered architecture (Presentation, Business and Data layers). Five components realise the requirements: the User Interface Component (Presentation Layer) used by Team Captains, Tournament Directors and Spectators; the Match & Standings Manager, Access Control Component and Bracket Generator Component (Business Layer); and the Database Component (Data Layer) holding teams, brackets, matches and standings. The Match & Standings Manager is the central orchestrator: it validates and records scores (FR-003), checks authorization with the Access Control Component before any restricted operation (NFR-002), asks the Bracket Generator Component to seed brackets and advance winners (FR-001, FR-002, FR-004), and recalculates standings against the Database Component (FR-005, NFR-001).

Style	Fit	Main reason
Layered	High	Requirements split cleanly into presentation, business (bracket/scoring/auth) and data concerns
Microservices	Low	FR-001–FR-005 form one tightly-coupled workflow; network hops would jeopardise the 200 ms NFR-001 target
Client-Server	Medium	Fits the client/server shape, but does not separate authorization, bracket logic and scoring the way the requirements demand

Two Reasons for the Choice

- **1. The requirements already separate into layers.** Bracket generation/seeding (FR-001, FR-002), score recording and tiebreaking (FR-003, UC-08), standings publication (FR-005) and access control (NFR-002) are distinct concerns that map directly onto the Business Layer's three components, while the UI serves three different actors (Team Captain, Tournament Director, Spectator) without needing to know how any of that logic works. This keeps each use case (UC-01 to UC-09) traceable to one component.
- **2. It keeps the 200 ms standings target achievable.** NFR-001 requires standings, goal differentials and the leaderboard to recalculate within 200 ms of a score submission. A layered, single-deployable application lets the Match & Standings Manager call the Database Component in-process, with no network hop — unlike a microservices split, which would add latency exactly where the requirement is strictest.

Security Advantage

All tournament-management operations (bracket generation, score submission, team/result administration) are required to pass through the Access Control Component before the Match & Standings Manager or Bracket Generator Component will act, directly satisfying NFR-002. The User Interface Component never talks to the Database Component directly, so a Spectator or unauthenticated request cannot reach match data or trigger a restricted operation, even though Spectators and Team Captains share the same UI as Tournament Directors.

Performance Benefit

Because the Match & Standings Manager, Bracket Generator, Access Control and Database components run in one deployable application, the score-submission path (FR-003 → FR-007 include "Update Standings" → bracket advancement) executes as local, in-process calls rather than network requests. This directly supports the 200 ms recalculation budget in NFR-001, which a distributed (microservices) design would put at risk.

Sports Tournament & Bracket Manager - Layered Architecture (UML Component Diagram)

